Linux Command Practice Report

Commands, screenshots and one-line explanations

Environment	Ubuntu Linux (QEMU virtual machine)
Shell	Bash
Date	05 November 2026

1. Creating and Renaming Files/Directories

Command: `mkdir test-dir`

Creates a new directory named test-dir.

```
yogesh@yogesh-QEMU-VM:~$ mkdir test-dir
```

Command: `cd test-dir`

Changes the current working directory to test-dir.

```
yogesh@yogesh-QEMU-VM:~$ cd test-dir
```

Command: `touch example.txt`

Creates an empty file named example.txt inside test-dir.

```
yogesh@yogesh-QEMU-VM:~/test-dir$ touch example.txt
```

Command: `mv example.txt renamed.txt`

Renames example.txt to renamed.txt using the move command.

```
yogesh@yogesh-QEMU-VM:~/test-dir$ mv example.txt renamed.txt
```

Command: `ls`

Lists the directory contents, confirming the file now appears as renamed.txt.

```
yogesh@yogesh-QEMU-VM:~/test-dir$ ls
renamed.txt
```

2. Viewing File Contents

Command: `cat /etc/passwd`

Displays the full contents of the /etc/passwd user account file (output truncated here).

```
yogesh@yogesh-QEMU-VM:~/test-dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
```

Command: `head -n 5 /etc/passwd`

Displays only the first 5 lines of /etc/passwd.

```
yogesh@yogesh-QEMU-VM:~/test-dir$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

Command: `tail -n 5 /etc/passwd`

Displays only the last 5 lines of /etc/passwd.

```
yogesh@yogesh-QEMU-VM:~/test-dir$ tail -n 5 /etc/passwd
polkitd:x:979:979:User for polkitd:/:/usr/sbin/nologin
rtkit:x:978:978:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:977:977:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
gdm:x:975:975:Gnome Display Manager:/var/lib/gdm3:/bin/false
yogesh:x:1000:1000:Yogesh:/home/yogesh:/bin/bash
```

3. Searching for Patterns

Command: `grep "root" /etc/passwd`

Finds and highlights every line in /etc/passwd that contains the word "root".

```
yogesh@yogesh-QEMU-VM:~/test-dir$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
nm-openvpn:x:984:984:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
```

4. Zipping and Unzipping

Command: `zip -r test-dirzip test-dir`

Recursively compresses the test-dir directory into a zip archive (test-dirzip.zip).

```
yogesh@yogesh-QEMU-VM:~$ zip -r test-dirzip test-dir
  adding: test-dir/ (stored 0%)
  adding: test-dir/renamed.txt (stored 0%)
```

Command: `mkdir unzipped_dir`

Creates a new directory named unzipped_dir to hold the extracted files.

```
yogesh@yogesh-QEMU-VM:~$ mkdir unzipped_dir
```

Command: `unzip test-dirzip.zip -d unzipped_dir`

Extracts the zip archive into the unzipped_dir directory.

```
yogesh@yogesh-QEMU-VM:~$ unzip test-dirzip.zip -d  unzipped_dir
Archive:  test-dirzip.zip
   creating: unzipped_dir/test-dir/
 extracting: unzipped_dir/test-dir/renamed.txt
```

5. Downloading Files

Command: `wget https://hetzner.de`

Downloads a web page from a URL; wget followed the 301 redirect and received 200 OK (saved as index.html).

```
yogesh@yogesh-QEMU-VM:~$ wget https://hetzner.de
--2026-11-05 09:10:16--  https://hetzner.de/
Resolving hetzner.de (hetzner.de)... 213.133.116.44, 2a01:4f8:0:1::7:4
Connecting to hetzner.de (hetzner.de)|213.133.116.44|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://www.hetzner.com/de/ [following]
--2026-11-05 09:10:17--  https://www.hetzner.com/de/
Resolving www.hetzner.com (www.hetzner.com)... 213.133.116.44, 2a01:4f8:0:1::7:4
Connecting to www.hetzner.com (www.hetzner.com)|213.133.116.44|:443... connected.
```

6. Changing Permissions

Command: `touch secure.txt`

Creates an empty file named secure.txt.

```
yogesh@yogesh-QEMU-VM:~$ touch secure.txt
```

Command: `chmod 444 secure.txt`

Sets secure.txt to read-only for owner, group and others.

```
yogesh@yogesh-QEMU-VM:~$ chmod 444 secure.txt
```

7. Working with Environment Variables

Command: `export Name="Hello, from Yogesh"`

Creates an environment variable and exports it so child processes can use it.

```
yogesh@yogesh-QEMU-VM:~$ export Name="Hello, from Yogesh"
```

Command: `echo $Name`

Prints the variable's value, confirming it was set correctly.

```
yogesh@yogesh-QEMU-VM:~$ echo $Name
Hello, from Yogesh
```